EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

Introduction: Equilibrium In Chemical Processes - Dynamic Equilibrium, Law Of Mass Action

1. In reversible chemical reaction equilibrium will establish when :

(A) Reactant is completely converted into product

(B) Rate of forward and backward reaction is equal

(C) Minimum yield of product

(D) Concentration of reactant and product is equal

CCE074

2. In which of the following reaction equilibrium cannot be established?

(A) $NaCl + AgNO_3 \longrightarrow AgCl + NaNO_3$

(B) $HCl + NaOH \longrightarrow NaCl + H_2O$

(C) $H_2SO_4 \longrightarrow 2H^{\oplus} + SO_4^{2\ominus}$

(D) All of these

CCE078

Law Of Chemical Equilibrium And Equilibrium Constant, Kp

3. The equilibrium concentration of B for the reversible reaction $A \rightleftharpoons B$ can be evaluated by the expression:

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(A) K_C (B) $\frac{k_f}{k_b}[A]_{eq}^{-1}$

(C) $\frac{k_f}{k_b}[A]_{eq}$ (D) $k_f k_b [A]_{eq}^{-1}$

CCE075

4. In a chemical equilibrium the equilibrium constant is found to be 2.5. If the rate constant of backward reaction is 10, the rate constant of forward reaction is –

(A) 2.5 (B) 0.25

(C) 25 (D) None

CCE077

Equilibrium Constant, Kc

5. 64 g of HI are present in a 2 litre vessel. The active mass of HI is:

(A) 0.5 (B) 0.25

(C) 1.0 (D) None of these

CCE079

6. When 4 moles of A are mixed with 4 moles of B, then 2 moles of C are formed at equilibrium, according to the reaction, $A + B \rightleftharpoons C + D$. The equilibrium constant is :-

(A) 4 (B) 1

(C) $\sqrt{2}$ (D) $\sqrt{4}$

CCE081

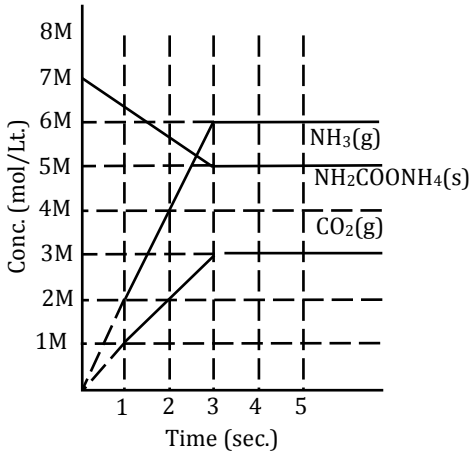
For More Join: @IITJEE_Advanced

7. Solid ammonium carbamate dissociates to give ammonia and carbon dioxide as follows

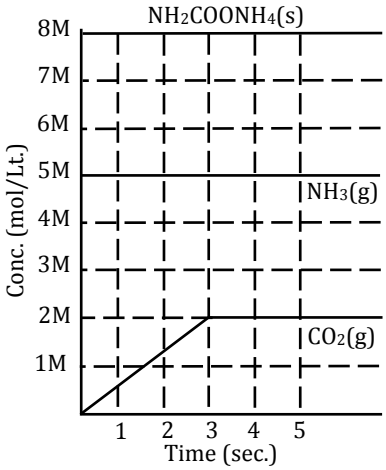
$NH_2COONH_4(s) \rightleftharpoons 2NH_3(g) + CO_2(g)$

which of the following graph correctly represents the equilibrium?

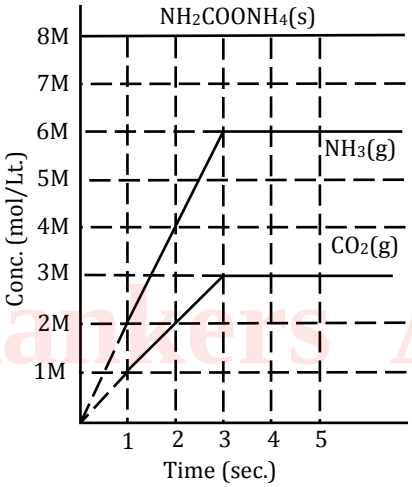
(A)



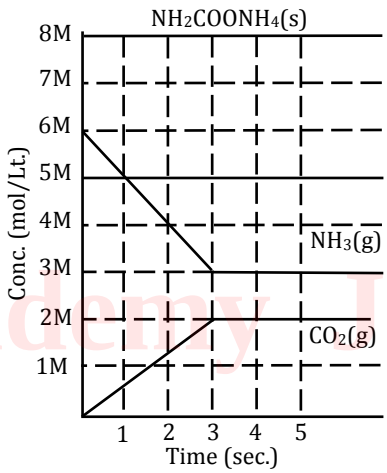
(B)



(C)



(D)



CCE082

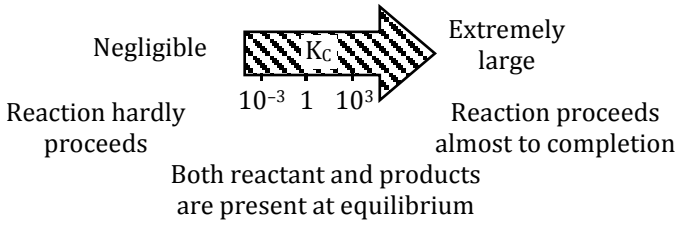
8. 5.1g NH_4SH is introduced in 3.0 L evacuated flask at 327°C. 30% of the solid NH_4SH decomposed to NH_3 and H_2S as gases. The K_P of the reaction at 327°C is (R = 0.082 L atm $mol^{-1}K^{-1}$, Molar mass of S = 32 g mol^{-1}, molar mass of N = 14g mol^{-1})

(A) 1×10^{-4} atm^2 (B) 4.9×10^{-3} atm^2 (C) 0.242 atm^2 (D) 0.242×10^{-4} atm^2

CCE091

Application Of Equilibrium Constant

9. Consider the following figure which shows dependence of extent of reaction on K_C.



Point out the correct statement(s) for the above diagram.

(A) If $K_C > 10^3$, products predominate over reactants.

(B) If $K_C < 10^{-3}$ reactants predominate over products.

(C) If K_C is in the range of 10^{-3} to 10^3, appreciable concentrations of both reactants and products are present.

(D) All of the above

CCE089

For More Join: @IITJEE_Advanced

10. A 10 L container at 300 K contains CO_2 gas at pressure of 0.2 atm and an excess solid CaO (neglect the volume of solid CaO). The volume of container is now decreased by moving the movable piston fitted in the container. What will be the maximum volume of container when pressure of CO_2 attains its maximum value given that $CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$ $K_p = 0.800$ atm

(A) 5 L
(B) 2.5 L
(C) 1 L
(D) The information is insufficient.

CCE098

Degree Of Dissociation

11. For the reaction: $2HI(g) \rightleftharpoons H_2(g) + I_2(g)$, the degree of dissociation (α) of HI(g) is related to equilibrium constant K_P by the expression.

(A) $\frac{1+2\sqrt{K_P}}{2}$ (B) $\sqrt{\frac{1+2K_P}{2}}$ (C) $\sqrt{\frac{2K_P}{1+2K_P}}$ (D) $\frac{2\sqrt{K_P}}{1+2\sqrt{K_P}}$

CCE088

12. In a 0.25 L tube dissociation of 4 mol of NO takes place. If its degree of dissociation is 10%. The value of K_P for reaction $2NO \rightleftharpoons N_2 + O_2$ is :

(A) $\frac{1}{(18)^2}$ (B) $\frac{1}{(8)^2}$
(C) $\frac{1}{16}$ (D) $\frac{1}{32}$

CCE085

Factors Affecting Equilibrium, Le- Chatelier's Principle, Problems On Le- Chatelier's Principle

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13. The role of catalyst in a chemical reaction is :-

(A) To help attain equilibrium in a shorter time.
(B) To lower the activation energy.
(C) To shift the equilibrium in such a way as to increase the concentration of the product
(D) Both 1 & 2

CCE084

14. In which of the following equilibrium reactions, the equilibrium would shift to right side, if total pressure is decreased :-

(A) $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$ (B) $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$
(C) $N_2O_4(g) \rightleftharpoons 2NO_2(g)$ (D) $H_2(g) + Cl_2(g) \rightleftharpoons 2HCl(g)$

CCE092

15. Does Le chatelier principle predict a change of equilibrium concentration for the following reaction if the gas mixture is compressed $N_2O_4(g) \rightleftharpoons 2NO_2(g)$

(A) Yes, backward reaction is favoured
(B) Yes, forward reaction is favoured
(C) No change
(D) No information

CCE095

16. In a vessel containing SO_3, SO_2 and O_2 at equilibrium, some helium gas is introduced so that the total pressure increases with temperature and volume remain constant. according to Le–chatelier principle, the dissociation of SO_3

(A) Increases
(B) Decreases
(C) Remains unaltered
(D) None of these

CCE097

For More Join: @IITJEE_Advanced

Simultaneous And Sequential Equilibrium

17. In a closed container following equilibrium will be attained –

$A(s) + B(g) \rightleftharpoons AB(g)$

$B(g) + C(g) \rightleftharpoons BC(g)$

On adding He gas (inert) to the above system at constant pressure & temperature

(A) Amount of AB(g) will be increased surely.

(B) Amount of B(g) will be decreased surely.

(C) Amount of C(g) will be decreased surely.

(D) Amount of BC(g) will be decreased surely.

CCE101

Relationship Between Equilibrium Constant K, Reaction Quotient Q And Gibb's Energy G

18. Which of the following is in favour of forward reaction :–

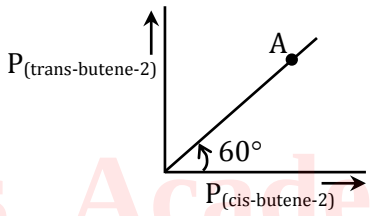
(A) $Q = K_C$ (B) $Q > K_C$

(C) $Q < K_C$ (D) None

CCE086

19. For the following isomerisation reaction

cis-butene-2 $\rightleftharpoons$ trans-butene-2; $K_P = 1.732$



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Which of the following statement is true at point 'A' ?

(A) $Q > K_P$ (B) $Q < K_P$

(C) $Q = K = 1$ (D) $Q = K = 1.732$

CCE090

For More Join: @IITJEE_Advanced

EXERCISE (O-2)

MULTIPLE CORRECT TYPE QUESTIONS

Law Of Chemical Equilibrium And Equilibrium Constant, K_p

1. What is/are true about equilibrium state?

(A) Catalyst has no effect on equilibrium state.

(B) Equilibrium constant is independent of initial concentration of reaction.

(C) The reaction ceases at equilibrium.

(D) Equilibrium constant is independent of pressure.

CCE102

Application Of Equilibrium Constant

2. In a vessel of 1.0 L capacity, $O_2(g)$ at 0.25 atm pressure and HCl(g) at 1.0 atm pressure are allowed to react in the presence of liquid water at 57oC.

$4HCl(g) + O_2(g) \rightleftharpoons 2Cl_2(g) + 2H_2O(g)$; $K_P = 5.0 \times 10^{12}$ atm^{-1}.

The volume occupied by liquid water is negligible, but it is sufficient at achieve equilibrium with water vapour. The vapour pressure of water at 57oC is 0.4 atm. Select the correct statement(s) regarding the equilibrium mixture.

(A) The partial pressure of water vapour at any stage of reaction is 0.4 atm.

(B) The partial pressure of $Cl_2(g)$ at equilibrium is 0.5 atm.

(C) The partial pressure of $O_2(g)$ at equilibrium is 5.0×10^{-4} atm.

(D) The partial pressure of HCl(g) at equilibrium is 2.0×10^{-3} atm.

Rankers Academy JEE

CCE109

Degree Of Dissociation

3. For dissociation of a gas N_2O_5 as $N_2O_5(g) \rightleftharpoons 2NO_2(g) + \frac{1}{2} O_2(g)$. The reaction is performed at constant temperature and volume. If D is the vapour density of equilibrium mixture, P_o is initial pressure of $N_2O_5(g)$ and M is molecular mass of N_2O_5, then the correct information(s) at the equilibrium is/are

(A) the total pressure of gases at equilibrium is $\frac{P_0 \cdot M}{2D}$

(B) the degree of dissociation of $N_2O_5(g)$ is $\frac{M-2D}{3D}$

(C) the partial pressure of $N_2O_5(g)$ at equilibrium $\frac{(5D-M)\cdot P_0}{3D}$

(D) the partial pressure of $O_2(g)$ at equilibrium is $\frac{(M-2D)\cdot P_0}{3D}$

CCE106

Factors Affecting Equilibrium, Le- Chatelier's Principle, Problems On Le-Chatelier's Principle

4. A chemical system is in equilibrium. Addition of a catalyst would result in

(A) increase in the rate of forward reaction.

(B) increase in the rate of reverse reaction.

(C) a new reaction pathway to reaction.

(D) increase the amount of heat evolved.

CCE111

For More Join: @IITJEE_Advanced

5. A cylinder fitted with a movable piston contains liquid water in equilibrium with water vapour at 25°C. Which operation results in a decrease in the equilibrium vapour pressure?

(A) Moving the piston downwards a short distance.

(B) Removing a small amount of vapour.

(C) Removing a small amount of the liquid water.

(D) Dissolving salt in the water.

CCE112

Relationship Between Equilibrium Constant K, Reaction Quotient Q And Gibb's Energy G

6. An amount of 1 mole of $PCl_3(g)$ and 1mole of $PCl_5(g)$ is taken in a vessel of 10L capacity maintained at 400 K. At equilibrium, the moles of Cl_2 is found to be 0.004.

(A) K_C for the reaction: $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$ is 0.0004 M.

(B) K_P for the reaction: $PCl_3(g) + Cl_2(g) \rightleftharpoons PCl_5(g)$ is $0.0004 \times (0.082 \times 400)$ atm.

(C) If $PCl_3(g)$ is added to the equilibrium mixture, ΔG at the new equilibrium becomes greater than the ΔG at old equilibrium.

(D) After equilibrium is achieved, the moles of PCl_3 are doubled and moles of Cl_2 are halved simultaneously, then the partial pressure of PCl_5 remains unchanged.

CCE103

MATRIX MATCH TYPE QUESTION

Factors Affecting Equilibrium, Le- Chatelier's Principle, Problems On Le- Chatelier's Principle

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7. Match the following :

	Column-I (Assume only reactant were present initially)		**Column-II**
(i)	For the equilibrium $NH_4I(s) \rightleftharpoons NH_3(g) + HI(g)$, if pressure is increased at equilibrium	(p)	Forward shift
(ii)	For the equilibrium $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$, volume is increased at equilibrium	(q)	No shift in equilibrium
(iii)	For the equilibrium $H_2O(g) + CO(g) \rightleftharpoons H_2(g) + CO_2(g)$, inert gas is added at constant pressure at equilibrium	(r)	Backward shift
(iv)	For the equilibrium $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$, Cl_2 is removed at equilibrium.	(s)	Final pressure is more than initial pressure

(A) (i - r); (ii - r); (iii - q); (iv - p)

(B) (i - p); (ii - q); (iii - r); (iv - s)

(C) (i - p); (ii - p); (iii - q); (iv - q)

(D) (i - p); (ii - q); (iii - r); (iv - r)

CCE099

For More Join: @IITJEE_Advanced

EXERCISE (S)

Part-1 : JEE Main (Numerical Value type)

- The answer to each question is a **Numerical Value**.
- For each question, enter the correct integer value (In case of non-integer value, the answer should be **rounded off** to the nearest Integer).

Law of Chemical Equilibrium and Equilibrium constant; Kp

1. A vessel at 1000 K contains CO_2 with a pressure of 0.5 atm. Some of the CO_2 is converted into CO on the addition of graphite. If the total pressure at equilibrium is 0.8 atm, the value of K_P is :

CCE025

Equilibrium constant; Kc

2. For the reaction $2SO_{3(g)} \rightleftharpoons 2SO_{2(g)} + O_{2(g)}$

Initially 5 moles of SO_3 is taken and at equilibrium 4 moles of SO_3 is dissociated, find out the value of K_C (in M) (Volume = 2litre)

CCE021

Relationship between Equilibrium constant K, Reaction Quotient Q and Gibb's Energy G

3. An equilibrium mixture of N_2, H_2 and NH_3 at 700 K contains 0.036 M N_2 and 0.15 M H_2. At this temperature, K_C for the reaction $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$ is 0.29. If concentration of NH_3 at equilibrium is 5.9×10^{-y} then calculate the value of 'y'.

CCE022

Rankers Academy JEE

Part-2 : JEE Advanced (Decimal type)

- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.

Law Of Chemical Equilibrium And Equilibrium Constant, K_p

4. Solid Ammonium carbamate dissociates as: $NH_2COONH_4\,(s) \rightleftharpoons 2NH_3(g) + CO_2(g)$. In a closed vessel solid ammonium carbamate is in equilibrium with its dissociation products. At equilibrium, ammonia is added such that the partial pressure of NH_3 at new equilibrium now equals the original total pressure. The ratio of total pressure at new equilibrium to that of original total pressure is x : 3, the value of 'x' is.

CCE136

5. In a vessel, two equilibrium are simultaneously established at same temperature as follows,

$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$ ...(i)

$N_2(g) + 2H_2(g) \rightleftharpoons N_2H_4(g)$ ...(ii)

Initially the vessel contains N_2 and H_2 in the molar ratio of 9 : 13. The equilibrium pressure is $7\,P_0$ in which pressure due to ammonia is P_0 and due to hydrogen is $2P_0$. Find the values of equilibrium constants (K_p's) for both the reactions.

CCE141

For More Join: @IITJEE_Advanced

Equilibrium Constant, Kc

6. For the reaction $A(g) + 2B(g) \rightleftharpoons C(g) + D(g)$; $K_c = 10^{12}$.

If the initial moles of A,B,C and D are 0.5, 1, 0.5 and 3.5 moles respectively in a one litre vessel. What is the equilibrium concentration of B (in millimole/litre) ?

CCE131

7. The ratio of equilibrium molar concentration of PCl_3 and PCl_5 at 300 K if initially 1 mol of PCl_5 is taken in a closed container of having volume 10 litre, is x : 1.00. The value of 'x' is.

[Given : $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$; $K_C = 10^{-2}$ at 300 K, $\sqrt{41} = 6.4$]

CCE132

8. At certain temperature (T) for the gas phase reaction

$2H_2O(g) + 2Cl_2(g) \rightleftharpoons 4HCl(g) + O_2(g)$ $\quad K_p = 12 \times 10^8$ atm

If Cl_2, HCl & O_2 are mixed in such a manner that the partial pressure of each is 2 atm and the mixture is brough into contact with excess of liquid water. What would be approximate partial pressure (in 10^{-3} atm) of Cl_2 when equilibrium is attained at temperature (T)?

[Given : Vapour pressure of water is 380 mm Hg at temperature (T)]

CCE138

Application Of Equilibrium Constant

9. Equilibrium constants are given (in atm) for the following reactions at 0° C:

$SrCl_2 \times 6H_2O(s) \rightleftharpoons SrCl_2 \times 2H_2O(s) + 4H_2O(g)$ $\quad K_p = 5 \times 10^{-12}$

$Na_2HPO_4 \times 12\,H_2O(s) \rightleftharpoons Na_2HPO_4 \times 7\,H_2O(s) + 5H_2O(g)$ $\quad K_p = 2.43 \times 10^{-13}$

$Na_2SO_4 \times 10\,H_2O(s) \rightleftharpoons Na_2SO_4(s) + 10\,H_2O(g)$ $\quad K_p = 1.024 \times 10^{-27}$

$$\left(\%\ \text{Relative humidity} = \frac{P_{H_2O}}{\text{Vapour pressure of } H_2O} \times 100\ \right)$$

The vapor pressure of water at 0°C is 4.56 torr.

Out of $SrCl_2 \times 2H_2O(s)$, $Na_2HPO_4 \times 7\,H_2O(s)$ and $Na_2SO_4(s)$,

Which is the most effective drying agent at 0°C?

CCE144

10. At certain temperature , $K_C = 1.0$ for reaction :

$NO_2(g) + NO(g) \rightleftharpoons N_2O(g) + O_2(g)$

equal moles of NO and NO_2 are to be placed in 5 litre container until N_2O concentration at equilibrium is 0.5M. How many mole of ($NO + NO_2$) must be placed in the container.

CCE145

Degree Of Dissociation

11. When 36.8g N_2O_4 (g) is introduced into a 1.0-litre flask at 27°C . The following equilibrium reaction occurs:

$N_2O_4(g) \rightleftharpoons 2NO_2(g)$; $K_p = 0.164$ atm.

(a) Calculate K_c of the equilibrium reaction. (in 10^{-3} M)

(b) What are the number of millimoles of N_2O_4 at equilibrium?

(c) What is the total pressure of gases (in atm) in the flask at equilibrium?

(d) What is the percent dissociation of N_2O_4?

CCE133

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12. One mole of N_2O_4 (g) at 300 K is left in a closed container under one atm. It is heated to 600 K when 20 % by mass of N_2O_4 (g) decomposes to NO_2 (g). Calculate resultant pressure (in atm).

CCE135

13. If vapour density of equilibrium mixture NO_2 and N_2O_4 is found to be 40 for the reaction,

$N_2O_4 \rightleftharpoons 2NO_2$. Find :

(a) Molar mass of equilibrium mixture (in gm/mol)
(b) Degree of dissociation
(c) Percentage of NO_2 in the mixture

CCE142

Factors Affecting Equilibrium, Le- Chatelier's Principle, Problems On Le- Chatelier's Principle

14. A definite amount of solid NH_4HS is placed in a flask already containing ammonia gas at a certain temperature and 0.50 atm pressure. NH_4HS decomposes to give NH_3 and H_2S and at equilibrium total pressure in flask is 0.84 atm. Calculate equilibrium constant for the reaction (in atm^2) :

CCE143

Simultaneous And Sequential Equilibrium

15. Two solids X and Y disssociate into gaseous products at a certain temperature as follows:
$X(s) \rightleftharpoons A(g) + C(g)$, and $Y(s) \rightleftharpoons B(g) + C(g)$. At a given temperature, pressure over excess solid X is 40 mm and total pressure over solid Y is 60 mm. When they are preset in separate containers. Calculate
(a) the values of K_p for two reactions (in mm^2)
(b) the ratio of moles of A and B in the vapour state over a mixture of X and Y.
(c) the total pressure of gases (in mm) over a mixture of X and Y.

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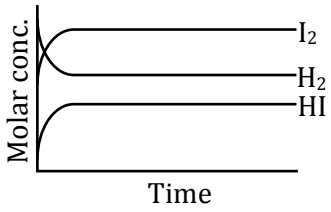
CCE139

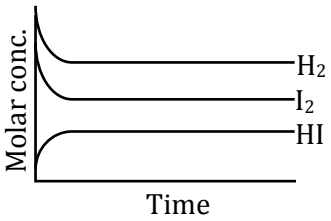
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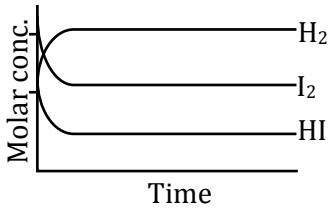
EXERCISE - JEE (Main) PYQ

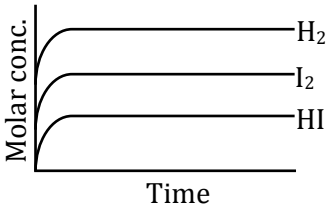
1. For the reaction,
$H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$

Attainment of equilibrium is predicted correctly by: **[JEE (Main) 2025]**

(1) 

(2) 

(3) 

(4) 

CCE199

2. A vessel at 1000 K contains CO_2 with a pressure of 0.5 atm. Some of CO_2 is converted into CO on addition of graphite. If total pressure at equilibrium is 0.8 atm, then K_P is : **[JEE (Main) 2025]**

(1) 0.18 atm (2) 1.8 atm
(3) 0.3 atm (4) 3 atm.

CCE200

3. 37.8 g N_2O_5 was taken in a 1 L reaction vessel and allowed to undergo the following reaction at 500 K
$2N_2O_{5(g)} \rightarrow 2N_2O_{4(g)} + O_{2(g)}$
The total pressure at equilibrium was found to be 18.65 bar.
Then, Kp = _____ $\times 10^{-2}$ [nearest integer]
Assume N_2O_5 to behave ideally under these conditions
Given : R = 0.082 bar L mol^{-1} K^{-1} **[JEE (Main) 2025]**

CCE201

4. The following concentrations were observed at 500 K for the formation of NH_3 from N_2 and H_2. At equilibrium:$[N_2] = 2 \times 10^{-2}$ M, $[H_2] = 3 \times 10^{-2}$ M and $[NH_3] = 1.5 \times 10^{-2}$M.
Equilibrium constant for the reaction is _____. **[JEE (Main) 2024]**

CCE162

5. 4.5 moles each of hydrogen and iodine is heated in a sealed ten litre vessel. At equilibrium, 3 moles of HI were found. The equilibrium constant for $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$ is ..... **[JEE (Main) 2023]**

CCE163

6. For the reaction $N_2O_4(g) \rightleftharpoons 2NO_2(g)$,K_p= 0.492 atm at 300K. K_c for the reaction at same temperature is__ $\times 10^{-2}$.
(Given : R = 0.082 L atm mol^{-1} K^{-1}) **[JEE (Main) 2024]**

CCE164

7. At –20 °C and 1 atm pressure, a cylinder is filled with equal number of H_2. I_2 and HI molecules for the reaction
$H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$, the K_P for the process is $x \times 10^{-1}$. x = _____.
[Given : R = 0.082 L atm K^{-1} mol^{-1}] **[JEE (Main) 2024]**
(1) 2 (2) 1 (3) 10 (4) 0.01

CCE165

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8. The ratio $\frac{K_P}{K_C}$ for the reaction:

$CO_{(g)} + \frac{1}{2}O_{2(g)} \rightleftharpoons CO_{2(g)}$ is: **[JEE (Main) 2024]**

(1) $(RT)^{1/2}$ (2) RT (3) 1 (4) $\frac{1}{\sqrt{RT}}$

CCE166

9. At 1990 K and 1 atm pressure, there are equal number of Cl_2 molecules and Cl atoms in the reaction mixture. The value K_P for the reaction $Cl_{2(g)} \rightleftharpoons 2Cl_{(g)}$ under the above conditions is $x \times 10^{-1}$. The value of x is _____.
(Rounded of to the nearest integer) **[JEE (Main) 2021]**

CCE047

10. $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$ **[JEE (Main) 2021]**

In an equilibrium mixture, the partial pressures are
$P_{SO_3} = 43$ kPa ; $P_{O_2} = 530$ Pa and $P_{SO_2} = 45$ kPa. The equilibrium constant K_P = __ × 10^{-2}. (Nearest integer)

CCE123

11. At 298 K

$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$; $K_1 = 4 \times 10^5$

$N_2(g) + O_2(g) \rightleftharpoons 2NO\,(g)$; $K_2 = 1.6 \times 10^{12}$

$H_2(g) + \frac{1}{2}O_2(g) \rightleftharpoons H_2O\,(g)$; $K_3 = 1.0 \times 10^{-13}$

Based on above equilibria, the equilibrium constant of the reaction,

$2NH_3(g) + \frac{5}{2}O_2(g) \rightleftharpoons 2NO\,(g) + 3H_2O\,(g)$

is _____ × 10^{-33}. (Nearest integer) **[JEE (Main) 2023]**

CCE126

12. Consider the reaction

$X_2Y(g) \rightleftharpoons X_2(g) + \frac{1}{2}Y_2(g)$

The equation representing correct relationship between the degree of dissociation (x) of $X_2Y(g)$ with its equilibrium constant Kp is_____.
Assume x to be very very small. **[JEE (Main) 2025]**

(1) $x = \sqrt[3]{\frac{2Kp}{p}}$ (2) $x = \sqrt[3]{\frac{2Kp^2}{p}}$

(3) $x = \sqrt[3]{\frac{Kp}{2p}}$ (4) $x = \sqrt[3]{\frac{Kp}{p}}$

CCE202

13. At temperature T, compound $AB_{2(g)}$ dissociates as $AB_{2(g)} \rightleftharpoons AB_{(g)} + \frac{1}{2}B_{2(g)}$ having degree of dissociation x (small compared to unity). The correct expression for x in terms of K_p and p is **[JEE (Main) 2025]**

(1) $\sqrt[3]{\frac{2K_p}{p}}$ (2) $\sqrt[4]{\frac{2K_p}{p}}$ (3) $\sqrt[3]{\frac{2K_p^2}{p}}$ (4) $\sqrt{K_p}$

CCE203

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14. (i) $X(g) \rightleftharpoons Y(g) + Z(g)$; $K_{p1} = 3$

(ii) $A(g) \rightleftharpoons 2B(g)$; $K_{p2} = 1$

If the degree of dissociation and initial concentration of both the reactants X(g) and A(g) are equal, then the ratio of the total pressure at equilibrium $\left(\frac{p_1}{p_2}\right)$ is equal to x : 1. The value of x is _____

(Nearest integer) **[JEE (Main) 2023]**

CCE128

15. For a concentrated solution of a weak electrolyte (K_{eq} = equilibrium constant) A_2B_3 of concentration 'c', the degree of dissociation 'α' is **[JEE (Main) 2023]**

(1) $\left(\frac{K_{eq}}{108c^4}\right)^{\frac{1}{5}}$ (2) $\left(\frac{K_{eq}}{6c^5}\right)^{\frac{1}{5}}$

(3) $\left(\frac{K_{eq}}{5c^4}\right)^{\frac{1}{5}}$ (4) $\left(\frac{K_{eq}}{25c^2}\right)^{\frac{1}{5}}$

CCE167

16. A mixture of 1 mole of H_2O and 1 mole of CO is taken in a 10 litre container and heated to 725 K. At equilibrium 40% of water by mass reacts with carbon monoxide according to the equation :

$CO(g) + H_2O(g) \rightleftharpoons CO_2(g) + H_2(g)$.

The equilibrium constant $K_C \times 10^2$ for the reaction is _____. (Nearest integer)

[JEE (Main) 2023]

CCE168

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17. $A_{(g)} \rightleftharpoons B_{(g)} + \frac{C}{2}_{(g)}$ The correct relationship between K_P, α and equilibrium pressure P is

[JEE (Main) 2024]

(1) $K_P = \frac{\alpha^{1/2} P^{1/2}}{(2+\alpha)^{1/2}}$ (2) $K_P = \frac{\alpha^{3/2} P^{1/2}}{(2+\alpha)^{1/2}(1-\alpha)}$

(3) $K_P = \frac{\alpha^{1/2} P^{3/2}}{(2+\alpha)^{3/2}}$ (4) $K_P = \frac{\alpha^{1/2} P^{1/2}}{(2+\alpha)^{3/2}}$

CCE169

18. Consider the equilibrium

$CO(g) + 3H_2(g) \rightleftharpoons CH_4(g) + H_2O(g)$

If the pressure applied over the system increases by two fold at constant temperature then

(A) Concentration of reactants and products increases.

(B) Equilibrium will shift in forward direction.

(C) Equilibrium constant increases since concentration of products increases.

(D) Equilibrium constant remains unchanged as concentration of reactants and products remain same.

Choose the **correct** answer from the options given below : **[JEE (Main) 2025]**

(1) (A) and (B) only (2) (A), (B) and (D) only

(3) (B) and (C) only (4) (A), (B) and (C) only

CCE204

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19. The equilibrium constant for the reaction

$$SO_3(g) \rightleftharpoons SO_2(g) + \frac{1}{2}O_2(g)$$

is $K_C = 4.9 \times 10^{-2}$. The value of K_C for the reaction given below is

$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$ is **[JEE (Main) 2024]**

(1) 4.9 (2) 41.6 (3) 49 (4) 416

CCE170

20. The following reaction occurs in the Blast furnace where iron ore is reduced to iron metal

$Fe_2O_{3(s)} + 3CO_{(g)} \rightleftharpoons Fe_{(l)} + 3CO_{2(g)}$

Using the Le-Chatelier's principle, predict which one of the following will not disturb the equilibrium. **[JEE (Main) 2024]**

(1) Addition of Fe_2O_3 (2) Addition of CO_2 (3) Removal of CO (4) Removal of CO_2

CCE171

21. For the given hypothetical reactions, the equilibrium constants are as follows:

$X \rightleftharpoons Y$; $K_1 = 1.0$

$Y \rightleftharpoons Z$; $K_2 = 2.0$

$Z \rightleftharpoons W$; $K_3 = 4.0$

The equilibrium constant for the reaction

$X \rightleftharpoons W$ is **[JEE (Main) 2024]**

(1) 6.0 (2) 12.0 (3) 8.0 (4) 7.0

CCE172

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22. Two solids dissociate as follows

$A(s) \rightleftharpoons B(g) + C(g)$; $K_{P_1} = x\ atm^2$

$D(s) \rightleftharpoons C(g) + E(g)$; $K_{P_2} = y\ atm^2$

The total pressure when both the solids dissociate simultaneously is :- **[JEE (Main) 2019]**

(1) $(x + y)$ atm (2) $x^2 + y^2$ atm

(3) $2\left(\sqrt{x+y}\right)$ atm (4) $\sqrt{x+y}$ atm

CCE043

23. Consider the following reaction approaching equilibrium at 27°C and 1 atm pressure

$$A + B \underset{K_r=10^2}{\overset{K_f=10^3}{\rightleftharpoons}} C + D$$

The standard Gibb's energy change $\left(\Delta_r G^\circ\right)$ at 27°C is (–) _____ kJ mol^{-1}

(Nearest integer).

(Given : R = 8.3 J K^{-1} mol^{-1} and ln 10=2.3) **[JEE (Main) 2023]**

CCE173

24. Ice and water are placed in a closed container at a pressure of 1 atm and temperature 273.15 K. If pressure of the system is increased 2 times, keeping temperature constant, then identify correct observation from following : **[JEE (Main) 2025]**

(1) Volume of system increases.

(2) Liquid phase disappears completely.

(3) The amount of ice decreases.

(4) The solid phase (ice) disappears completely.

CCE205

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EXERCISE - JEE (Advanced) PYQ

1. The thermal dissociation equilibrium of $CaCO_3(s)$ is studied under different conditions.

$CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$

For this equilibrium, the correct statement(s) is(are) **[JEE (Advanced) 2013]**

(A) ΔH is dependent on T

(B) K is independent of the initial amount of $CaCO_3$

(C) K is dependent on the pressure of CO_2 at a given T

(D) ΔH is independent of the catalyst if any

CCE072

2. For the following reaction, the equilibrium constant K_C at 298 K is 1.6×10^{17}.

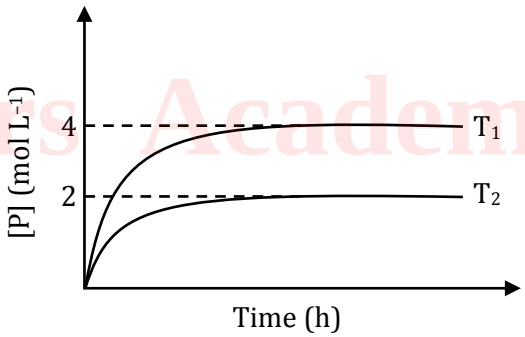
$Fe^{2\oplus}(aq) + S^{2\ominus}(aq) \rightleftharpoons FeS(s)$

When equal volumes of 0.06 M $Fe^{2\oplus}(aq)$ and 0.2 M $S^{2\ominus}(aq)$ solutions are mixed, the equilibrium concentration of $Fe^{2\oplus}(aq)$ is found to be $Y \times 10^{-17}$ M. The value of Y is (upto two decimal places.)

[JEE (Advanced) 2019]

CCE073

3. In a one-litre flask, 6 moles of A undergoes the reaction A (g) $\rightleftharpoons$ P (g). The progress of product formation at two temperatures (in Kelvin), T_1 and T_2, is shown in the figure:



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If $T_1 = 2T_2$ and $(\Delta G_2^{\ominus} - \Delta G_1^{\ominus}) = RT_2 \ln x$, then the value of x is _____.

[$\Delta G_1^{\ominus}$ and $\Delta G_2^{\ominus}$ are standard Gibb's free energy change for the reaction at temperatures T_1 and T_2, respectively.]

CCE130

[JEE (Advanced) 2023]

Common content for Q. 4 to 5

Thermal decomposition of gaseous X_2 to gaseous X at 298 K takes place according to the following equation :

$$X_2(g) \rightleftharpoons 2X(g)$$

The standard reaction Gibbs energy, $\Delta_r G°$, of this reaction is positive. At the start of the reaction, there is one mole of X_2 and no X. As the reaction proceeds, the number of moles of X formed is given by β. Thus, $\beta_{equilibrium}$ is the number of moles of X formed at equilibrium. The reaction is carried out at a constant total pressure of 2 bar. Consider the gases to behave ideally.

(Given : R = 0.083 L bar K^{-1} mol^{-1})

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4. The equilibrium constant K_P for this reaction at 298 K, in terms of $\beta_{equilibrium}$, is **[JEE (Advanced) 2016]**

(A) $\frac{8\beta^2_{equilibrium}}{2-\beta_{equilibrium}}$ (B) $\frac{8\beta^2_{equilibrium}}{4-\beta^2_{equilibrium}}$ (C) $\frac{4\beta^2_{equilibrium}}{2-\beta_{equilibrium}}$ (D) $\frac{4\beta^2_{equilibrium}}{4-\beta^2_{equilibrium}}$

CCE174

5. The **INCORRECT** statement among the following for this reaction is **[JEE (Advanced) 2016]**

(A) Decrease in the total pressure will result in formation of more moles of gaseous X

(B) At the start of the reaction, dissociation of gaseous X_2 takes place spontaneously

(C) $\beta_{equilibrium} = 0.7$

(D) $K_C < 1$

CCE175

6. The standard state Gibbs free energies of formation of C(graphite) and C(diamond) at T = 298 K are

$\Delta_f G^\circ$ [C(graphite)] = 0 kJ mol^{-1}

$\Delta_f G^\circ$ [C(diamond)] = 2.9 kJ mol^{-1}

The standard state means that the pressure should be 1 bar, and substance should be pure at a given temperature. The conversion of graphite [C(graphite)] to diamond [C(diamond)] reduces its volume by 2×10^{-6} m^3 mol^{-1}. If C(graphite) is converted to C(diamond) isothermally at T = 298 K, the pressure at which C(graphite) is in equilibrium with C(diamond), is

[Useful information : 1 J = 1 kg m^2 s^{-2} ; 1 Pa = 1 kg m^{-1} s^{-2} ; 1 bar = 10^5 Pa]

[JEE (Advanced) 2017]

(A) 14501 bar (B) 29001 bar (C) 58001 bar (D) 1405 bar

CCE176

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7. For a reaction taking place in a container in equilibrium with its surroundings, the effect of temperature on its equilibrium constant K in terms of change in entropy is described by **[JEE (Advanced) 2017]**

(A) With increase in temperature, the value of K for exothermic reaction decreases because the entropy change of the system is positive.

(B) With increase in temperature, the value of K for endothermic reaction increases because unfavourable change in entropy of the surroundings decreases.

(C) With increase in temperature, the value of K for exothermic reaction decreases because favourable change in entropy of the surroundings decreases.

(D) With increase in temperature, the value of K for endothermic reaction increases because the entropy changes of the system negative.

CCE177

8. The surface of copper gets tarnished by the formation of copper oxide. N_2 gas was passed to prevent the oxide formation during heating of copper at 1250 K. However, the N_2 gas contains 1 mole % of water vapour as impurity. The water vapour oxidises copper as per the reaction given below

$2Cu(s) + H_2O(g) \rightarrow Cu_2O(s) + H_2(g)$

p_{H2} is the minimum partial pressure of H_2 (in bar) needed to prevent the oxidation at 1250 K. The value of $\ln(P_{H_2})$ is _____.

(Given : total pressure = 1 bar, R (universal gas constant) = 8 $JK^{-1}mol^{-1}$, ln(10) = 2.3. Cu(s) and $Cu_2O(s)$ are mutually immiscible.

At 1250 K : $2Cu(s) + 1/2O_2(g) \rightarrow Cu_2O(s)$; ΔG^0 = – 78,000 J mol^{-1}

$H_2(g) + 1/2O_2(g) \rightarrow H_2O(g)$; ΔG^0 =–1,78,000 J mol^{-1};G is the Gibbs energy)

[JEE (Advanced) 2018]

CCE178

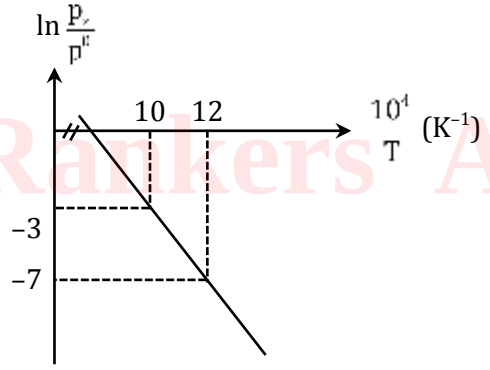
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9. Consider the following reversible reaction,
$A(g) + B(g) \rightleftharpoons AB(g)$.
The activation energy of the backward reaction exceeds that of the forward reaction by 2RT (in J mol^{-1}). If the pre-exponential factor of the forward reaction is 4 times that of the reverse reaction, the absolute value of $\Delta°G$ (in J mol^{-1}) for the reaction at 300 K is___.
(Given ; ln (2) = 0.7, RT = 2500 J mol^{-1} at 300 K and G is the Gibbs energy) **[JEE (Advanced) 2018]**
CCE179

10. Consider the reaction $A \rightleftharpoons B$ at 1000 K. At time t', the temperature of the system was increased to 2000 K and the system was allowed to reach equilibrium. Throughout this experiment the partial pressure of A was maintained at 1 bar. Given below is the plot of the partial pressure of B with time. What is the ratio of the standard Gibbs energy of the reaction at 1000 K to that at 2000 K?
[JEE (Advanced) 2020]
CCE180

Common content for Q. 11 - 12

For the reaction $X(s) \rightleftharpoons Y(s) + Z(g)$, the plot of $\ln \frac{p_z}{p^0}$ versus $\frac{10^4}{T}$ is given below (in solid line), where p_z is the pressure (in bar) of the gas Z at temperature T and P^1 =1 bar.



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(Given, $\frac{d(\ln K)}{d\left(\frac{1}{T}\right)} = -\frac{\Delta H^1}{R}$ where the equilibrium constant, $K = \frac{p_z}{p^0}$ and the gas constant, R = 8.314 J K^{-1} mol^{-1})

11. The value of standard enthalpy, ΔH^0 (in kJ mol^{-1}) for the reaction is____ . **[JEE (Advanced) 2021]**
CCE181

12. The value of ΔS^o (in J K^{-1} mol^{-1}) for the given reaction, at 1000 K is_____. **[JEE (Advanced) 2021]**
CCE182

For More Join: @IITJEE_Advanced

ANSWER KEY

EXERCISE (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	B	D	C	C	B	B	C	C	D	B
Que.	11	12	13	14	15	16	17	18	19	
Ans.	D	A	D	C	A	C	D	C	D	

EXERCISE (O-2)

Que.	1	2	3	4	5	6	7	
Ans.	ABD	ABCD	ABC	ABC	D	AD	A	

EXERCISE (S)

1. 2 **2.** 16 **3.** 3 **4.** 3.44

5. $K_{P_1} = \frac{1}{20P_0^2}$ **6.** 0.20 **7.** 0.36 or 0.37 **8.** 3.60

9. $(SrCl_2 . 2H_2O)$ **10.** 10.00

11. (a) 6.66 or 0.67, (b) 375.00, (c) 10.45 or 10.46, (d) 6.25 **12.** 2.40

13. (a) 80, (b) 0.15, (c) 26.08 or 26.09 **14.** 0.11

15. (a) 400, 900, (b) 4 : 9 , (c) 72.15

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EXERCISE - JM PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	2	2	962	417	1	2	3	4	5	172
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	4	2	3	12	1	44	2	1	4	1
Que.	21	22	23	24						
Ans.	3	3	6	4						

EXERCISE - JA PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	ABD	8.92 or 8.93	8	B	C	A	BC	–14.6	8500	0.25
Que.	11	12								
Ans.	166.28	141.33								

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